

Mathematical Formulation of the Proactive Golden-Hour Bridge Model (PGBM) for Post-Earthquake UAV-Assisted Response

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Abstract—This document presents the formal mathematical formulation of the Proactive Golden-Hour Bridge Model (PGBM), a UAV-assisted emergency response system for post-earthquake scenarios. The system operates as an online decision-making framework where Responder UAVs are dynamically assigned to detected survivors to provide immediate assistance during the critical "Golden Hour" before ground rescue teams physically arrive. The model captures the coupling between survivor task states, UAV dynamics, battery constraints, time-dependent urgency decay, and 3D routing feasibility. The global system state is formally defined, and the event-triggered decision policy is mathematically characterized.

I. INTRODUCTION

In the aftermath of a major earthquake, Scout UAVs can rapidly detect trapped survivors using thermal and optical sensors. However, Ground Rescue Teams face significant delays due to collapsed infrastructure and blocked roads, creating a deadly gap between detection and physical rescue. The Proactive Golden-Hour Bridge Model (PGBM) introduces an intermediate UAV-assisted response layer where Responder UAVs are dispatched immediately to provide emergency supplies, two-way communication, and GPS relay before ground teams arrive.

A. System Overview

The PGBM consists of four primary actors:

- 1) **Scout UAVs:** Detect survivors and transmit detection packets.
- 2) **Command Center:** Maintains global state and executes online allocation/routing decisions.
- 3) **Responder UAVs:** Execute bridge missions to deliver aid.
- 4) **Ground Rescue Team:** Performs final physical extraction (outside core model).

B. Core Assumptions

- Obstacles are static during a single mission (Version 1).
- Communication between all actors is reliable.
- Each Responder UAV serves one survivor at a time.
- Survivor tasks are independent and non-preemptive (baseline).

II. MATHEMATICAL NOTATION

Throughout this formulation, bold uppercase letters denote sets, bold lowercase denote vectors, and scalar values use standard notation. The time variable t denotes the current decision time (continuous). All decisions are event-triggered.

III. SYSTEM ENTITIES AND STATE DEFINITIONS

A. Survivor Task Model

Each survivor detection is modeled as a task $i \in \mathcal{S}(t)$, where $\mathcal{S}(t)$ is the set of all known survivor tasks at time t . Survivor i is defined as:

$$S_i = (x_i, y_i, z_i, t_i^{\text{det}}, c_i, \sigma_i, u_i(t), a_i, \text{assigned_to}_i) \quad (1)$$

with the following parameters:

TABLE I
SURVIVOR TASK PARAMETERS

Parameter	Symbol	Domain
Location	(x_i, y_i, z_i)	$\mathbb{R}^3$
Detection Time	t_i^{det}	$\mathbb{R}^+$
Confidence Score	c_i	$[0, 1]$
Severity	σ_i	$[0, 1]$
Urgency	$u_i(t)$	$[0, 1]$
Assignment Flag	a_i	$\{0, 1\}$
Assigned UAV	assigned_to_i	$\mathcal{D}(t) \cup \{\emptyset\}$

Severity σ_i denotes the static medical criticality at detection (independent of time), with $\sigma_i = 1$ representing the most critical condition. Two possible scenarios for obtaining σ_i are: (1) *Scout AI Inference*: estimated automatically from Scout UAV thermal/optical imagery and rubble context via a classification model (e.g., motion/heat signature analysis), and (2) *Human-in-the-Loop*: tagged manually by the Command Center operator reviewing the Scout video feed. If no severity estimate is available, $\sigma_i = 1$ uniformly.

B. Urgency Decay Function

The urgency of survivor i decays over time according to the Golden Hour principle. The primary model (Version 1) is exponential decay:

$$u_i(t) = \sigma_i \cdot e^{-\alpha(t-t_i^{\text{det}})} \quad (2)$$

where $\alpha > 0$ is the decay rate constant with $\alpha = \ln 2/T_{\text{half}}$ and T_{half} is the half-life of urgency (e.g., 30 minutes), and σ_i is the severity at detection so that $u_i(t_i^{\text{det}}) = \sigma_i$. This captures that a survivor found 10 minutes ago has higher salvageability than one found 90 minutes ago at the same current time t , with value decaying as waiting time increases.

Alternative (Remark): A logistic decay model can capture the "critical threshold" phenomenon:

$$u_i(t) = \sigma_i \cdot \frac{1}{1 + e^{k(t - t_i^{\text{det}} - T_{\text{half}})}} \quad (3)$$

where k controls the steepness of the decay. This alternative is deferred to future work for injury-specific cliff modeling.

C. Confidence Score and Verification Policy

Each detection carries a confidence score $c_i \in [0, 1]$ representing the Scout UAV detector's estimated probability that the detection is a true survivor (e.g., thermal/optical AI output). A threshold $c_{\text{thresh}} \in [0, 1]$ (e.g., 0.5) separates high- and low-confidence regimes:

$$\text{eligible}(i) = \begin{cases} 1 & \text{if } c_i \geq c_{\text{thresh}} \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

Detections with $c_i \geq c_{\text{thresh}}$ enter the high-confidence queue $\mathcal{Q}_{\text{high}}(t)$ and are eligible for Responder assignment. Detections with $c_i < c_{\text{thresh}}$ enter the low-confidence queue $\mathcal{Q}_{\text{low}}(t)$ and are *not* eligible for Responder dispatch; they require verification (Scout re-scan or operator review) before promotion to $\mathcal{Q}_{\text{high}}$. In Version 1, $\mathcal{Q}_{\text{low}}$ is modeled as a holding set and verification is abstracted as an exogenous promotion event. The offloading ILP is defined only over $\mathcal{Q}_{\text{high}}(t)$.

D. Responder UAV Model

Each Responder UAV $j \in \mathcal{D}(t)$ is defined as:

$$D_j = (x_j, y_j, z_j, p_j^{\text{base}}, B_j, A_j, s_j, \text{status}_j) \quad (5)$$

with the following parameters:

TABLE II
RESPONDER UAV PARAMETERS

Parameter	Symbol	Domain
Position	(x_j, y_j, z_j)	$\mathbb{R}^3$
Base Station	p_j^{base}	$\mathbb{R}^3$
Remaining Battery	B_j	$\mathbb{R}^+$
Availability	A_j	$\{0, 1\}$
Current Task	s_j	$\mathcal{S}(t) \cup \{\emptyset\}$
Mission Status	status_j	$\{\text{Idle}, \text{EnRoute}, \text{AtSurvivor}, \text{Returning}\}$

E. Availability States

The availability A_j is derived from the UAV's status:

$$A_j = \begin{cases} 1 & \text{if } \text{status}_j = \text{Idle} \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

F. Environment Model

The disaster environment is represented as:

$$\mathcal{E} = (V, O) \quad (7)$$

where:

- $V \subset \mathbb{R}^3$: Navigable free space
- $O \subset \mathbb{R}^3$: Static obstacle set (buildings, rubble, no-fly zones)

Note: The hazard map $R : V \rightarrow [0, 1]$ is deferred to the Routing Problem formulation.

G. Task Queue

The ordered queue of unassigned survivors is defined as a derived view:

$$\mathcal{Q}(t) = \text{order}(\{S_i \in \mathcal{S}(t) \mid a_i = 0\}) \quad (8)$$

ordered by priority:

$$\text{priority}(i) = \beta_1 \cdot u_i(t) + \beta_2 \cdot c_i - \beta_3 \cdot (t - t_i^{\text{det}}) \quad (9)$$

Note: $\mathcal{Q}(t)$ is not an independent state component but a derived view of $\mathcal{S}(t)$ for scheduling. With verification threshold c_{thresh} , the queue partitions as:

$$\mathcal{Q}_{\text{high}}(t) = \{S_i \in \mathcal{Q}(t) \mid c_i \geq c_{\text{thresh}}\}, \quad \mathcal{Q}_{\text{low}}(t) = \{S_i \in \mathcal{Q}(t) \mid c_i < c_{\text{thresh}}\} \quad (10)$$

Only $\mathcal{Q}_{\text{high}}(t)$ is eligible for the offloading assignment; $\mathcal{Q}_{\text{low}}(t)$ is held for verification. Note: $\text{priority}(i)$ is a pre-filter ordering heuristic for $\mathcal{Q}_{\text{high}}(t)$ (e.g., for greedy fallback or tie-breaking); the optimal assignment is determined by the ILP cost $C_{ij}(t)$, not by priority order.

IV. DECISION VARIABLES

A. Assignment Variable

$$x_{ij}(t) \in \{0, 1\} \quad (11)$$

$$x_{ij}(t) = \begin{cases} 1 & \text{if UAV } j \text{ is assigned to survivor } i \text{ at time } t \\ 0 & \text{otherwise} \end{cases} \quad (12)$$

B. Route Variable

The route $P_{ij}(t)$ is a sequence of waypoints from UAV j 's current position to survivor i :

$$P_{ij} = \{p_0, p_1, \dots, p_K\} \quad (13)$$

where:

- $p_0 = (x_j, y_j, z_j)$: Current UAV position
- $p_K = (x_i, y_i, z_i)$: Survivor location
- $p_k \in V$ for all k : All waypoints in navigable space

V. COST FUNCTIONS

A. Travel Time

$$T_{ij} = T(P_{ij}) = \sum_{k=1}^K \frac{\|p_k - p_{k-1}\|}{v_{\max}} \quad (14)$$

where $v_{\max}$ is the maximum UAV speed.

B. Energy Consumption

$$E_{ij} = E(P_{ij}) = \sum_{k=1}^K (e_{\text{travel}} \cdot \|p_k - p_{k-1}\| + e_{\text{vert}} \cdot \Delta z_k) \quad (15)$$

where:

- e_{travel} : Energy per unit horizontal distance
- e_{vert} : Energy penalty for vertical climb
- $\Delta z_k = \max(0, z_k - z_{k-1})$: Vertical gain

C. Service-Related Energy

If the UAV hovers at the survivor location for duration T_{service} (V1 fixed parameter, e.g., $T_{\text{service}} = 300$ s):

$$E_{\text{hover}} = e_{\text{hover}} \cdot T_{\text{service}} \quad (16)$$

D. Return Energy

The energy to return from survivor i to the base of UAV j is:

$$E_{\text{return}}(i, j) = E(P_{i \rightarrow \text{base}(j)}) = \sum_{k=1}^{K'} (e_{\text{travel}} \cdot \|p_k - p_{k-1}\| + e_{\text{vert}} \cdot \Delta z_k) \quad (17)$$

where $P_{i \rightarrow \text{base}(j)} = \{p_0 = p_i, \dots, p_{K'} = p_j^{\text{base}}\}$ is the feasible route from p_i to p_j^{base} with $p_k \in V$. This explicitly uses p_j^{base} from the UAV model and is generally *not* equal to $E(P_{ij})$ due to differing start position $p_j \neq p_j^{\text{base}}$ during online operation and asymmetric vertical cost.

E. Total Energy Required

The worst-case (conservative) total energy for a roundtrip mission is:

$$E_{ij}^{\text{total}} = E(P_{ij}) + E_{\text{hover}} + E_{\text{return}}(i, j) \quad (18)$$

where $P_{ij} = P_{j \rightarrow i}$ is the outbound route from current position p_j to survivor p_i . Variant for landing at survivor (no return): $E_{ij}^{\text{oneway}} = E(P_{ij}) + E_{\text{hover}}$, deferred to future work; V1 assumes roundtrip.

F. Assignment Cost

The cost of assigning UAV j to survivor i is:

$$C_{ij}(t) = \alpha \cdot T_{ij} + \beta \cdot E_{ij}^{\text{total}} - \gamma \cdot u_i(t) \cdot c_i \quad (19)$$

where $\alpha, \beta, \gamma > 0$ are weighting coefficients.

VI. FEASIBILITY CONSTRAINTS

A. Battery Constraint

A UAV j can be assigned to survivor i only if:

$$E_{ij}^{\text{total}} + E_{\text{reserve}} \leq B_j(t) \quad (20)$$

where $E_{\text{reserve}} = 0.2 \cdot B_{\max}$ is a safety reserve (20% of maximum battery capacity $B_{\max}$), $B_j(t) \in [0, B_{\max}]$ is the remaining battery, and $B_{\max}, T_{\text{service}}, e_{\text{travel}}, e_{\text{vert}}, e_{\text{hover}}, v_{\max}$ are global V1 constants.

B. Reachability Constraint

A route P_{ij} exists only if:

$$p_k \in V \quad \forall k \in \{0, 1, \dots, K\} \quad (21)$$

i.e., the entire path lies in navigable space. Reachability is a boolean oracle provided by the Routing Problem (e.g., 3D A* over $V \setminus O$); offloading consumes exists(P_{ij}) without defining discretization.

C. Availability Constraint

$$x_{ij}(t) \leq A_j \quad (22)$$

D. Feasible Assignment Condition

A complete feasibility check is:

$$\text{Feasible}(i, j) = (A_j = 1) \wedge (c_i \geq c_{\text{thresh}}) \wedge (P_{ij} \text{ exists}) \wedge (E(P_{j \rightarrow i}) + E_{\text{hover}} + E_{\text{return}}(i, j) \leq B_j(t)) \quad (23)$$

VII. ASSIGNMENT CONSTRAINTS

A. Single Assignment per Survivor

Each survivor is assigned to at most one Responder UAV:

$$\sum_{j=1}^M x_{ij}(t) \leq 1 \quad \forall i \in \mathcal{S}(t) \quad (24)$$

B. Single Task per UAV

Each UAV is assigned to at most one survivor at a time:

$$\sum_{i=1}^{N(t)} x_{ij}(t) \leq 1 \quad \forall j \in \mathcal{D}(t) \quad (25)$$

C. Feasibility Enforcement

$$x_{ij}(t) = 0 \quad \text{if } \neg \text{Feasible}(i, j) \quad (\text{includes } c_i < c_{\text{thresh}}) \quad (26)$$

D. Assignment to Available UAVs Only

$$x_{ij}(t) \leq A_j \quad \forall i, j \quad (27)$$

VIII. OBJECTIVE FUNCTION

A. Primary Objective: Minimize Cost

$$\min_{\mathbf{x}(t)} \sum_{i=1}^{N(t)} \sum_{j=1}^M x_{ij}(t) \cdot C_{ij}(t) \quad (28)$$

B. Alternative: Maximize Value

$$\max_{\mathbf{x}(t)} \sum_{i=1}^{N(t)} \sum_{j=1}^M x_{ij}(t) \cdot u_i(t) \cdot c_i \quad (29)$$

subject to:

$$\sum_{j=1}^M x_{ij}(t) \leq 1 \quad \forall i \quad (30)$$

$$\sum_{i=1}^{N(t)} x_{ij}(t) \leq 1 \quad \forall j \quad (31)$$

$$x_{ij}(t) \in \{0, 1\} \quad (32)$$

$$x_{ij}(t) = 0 \quad \text{if } \neg \text{Feasible}(i, j) \quad (33)$$

IX. GLOBAL SYSTEM STATE

A. Complete State Definition

The global system state at time t is:

$$X(t) = (\mathcal{S}(t), \mathcal{D}(t), \mathcal{E}(t), \mathcal{Q}(t)) \quad (34)$$

where:

- $\mathcal{S}(t)$: Set of survivor tasks with their complete state
- $\mathcal{D}(t)$: Set of Responder UAVs with their complete state
- $\mathcal{E}(t)$: Environment state
- $\mathcal{Q}(t)$: Ordered task queue of unassigned survivors

B. State Component Details

1) Survivor State Set:

$$\mathcal{S}(t) = \{S_i \mid i = 1, \dots, N(t)\} \quad (35)$$

where:

$$S_i = (x_i, y_i, z_i, t_i^{\text{det}}, c_i, \sigma_i, u_i(t), a_i, \text{assigned_to}_i) \quad (36)$$

2) UAV State Set:

$$\mathcal{D}(t) = \{D_j \mid j = 1, \dots, M\} \quad (37)$$

where:

$$D_j = (x_j, y_j, z_j, p_j^{\text{base}}, B_j, A_j, s_j, \text{status}_j) \quad (38)$$

3) Environment State:

$$\mathcal{E}(t) = (V, O(t)) \quad (39)$$

For Version 1, obstacles are static: $O(t) = O$, thus $\mathcal{E}(t) = \mathcal{E}$. Hazard map $R(t)$ is deferred to routing.

4) Task Queue:

$\mathcal{Q}(t) = \text{order}([S_i \in \mathcal{S}(t) \mid a_i = 0])$ derived view ordered by priority (40)
with partition $\mathcal{Q}_{\text{high}}(t) = \{i \in \mathcal{Q}(t) \mid c_i \geq c_{\text{thresh}}\}$ and $\mathcal{Q}_{\text{low}}(t) = \{i \in \mathcal{Q}(t) \mid c_i < c_{\text{thresh}}\}$; only $\mathcal{Q}_{\text{high}}$ is offloading-eligible.

X. ONLINE DECISION PROBLEM

A. Decision Policy

At each event time t , the Command Center computes:

$$\pi(t) = f(X(t)) = (\mathbf{x}(t), \mathbf{P}(t)) \quad (41)$$

where:

- $\mathbf{x}(t) = [x_{ij}(t)]$: Assignment matrix of size $N(t) \times M$
- $\mathbf{P}(t) = [P_{ij}(t)]$: Route matrix for all assignments

B. Event-Triggered Recalculation

The system is online because:

$$\mathcal{S}(t_1) \neq \mathcal{S}(t_2) \quad (\text{new detections}) \quad (42)$$

$$\mathcal{D}(t_1) \neq \mathcal{D}(t_2) \quad (\text{UAV state changes}) \quad (43)$$

$$u_i(t_1) \neq u_i(t_2) \quad (\text{urgency decay}) \quad (44)$$

C. Recalculation Triggers

The decision $\pi(t)$ is recomputed at:

- 1) **New Survivor Detection**: $\mathcal{S}(t) \leftarrow \mathcal{S}(t) \cup \{S_{\text{new}}\}$
- 2) **UAV Becomes Available**: $A_j = 0 \rightarrow A_j = 1$
- 3) **Periodic Re-evaluation**: Every ΔT seconds (optional)

D. State Transition Functions

1) New Detection:

$$\mathcal{S}(t^+) = \mathcal{S}(t^-) \cup \{S_{\text{new}}\} \quad (45)$$

$$\mathcal{Q}(t^+) = \mathcal{Q}(t^-) \cup \{S_{\text{new}}\} \quad (46)$$

$$N(t^+) = N(t^-) + 1 \quad (47)$$

2) UAV Dispatch:

$$D_j : A_j \leftarrow 0 \quad (48)$$

$$D_j : s_j \leftarrow S_i \quad (49)$$

$$D_j : \text{status}_j \leftarrow \text{EnRoute} \quad (50)$$

$$S_i : a_i \leftarrow 1 \quad (51)$$

$$S_i : \text{assigned_to}_i \leftarrow j \quad (52)$$

$$\mathcal{Q}(t) \leftarrow \mathcal{Q}(t) \setminus \{S_i\} \quad (53)$$

3) UAV Reaches Survivor:

$$D_j : \text{status}_j \leftarrow \text{AtSurvivor} \quad (54)$$

4) UAV Completes Mission:

$$D_j : A_j \leftarrow 1 \quad (55)$$

$$D_j : s_j \leftarrow \emptyset \quad (56)$$

$$D_j : \text{status}_j \leftarrow \text{Idle} \quad (57)$$

$$B_j \leftarrow B_j - E_{ij}^{\text{consumed}} \quad (58)$$

XI. ROUTING MODEL

A. Path Definition

A route P_{ij} is a sequence of $K + 1$ waypoints:

$$P_{ij} = \{p_0, p_1, \dots, p_K\} \quad (59)$$

with:

$$p_0 = (x_j, y_j, z_j) \quad (60)$$

$$p_K = (x_i, y_i, z_i) \quad (61)$$

$$p_k \in V \quad \forall k \quad (62)$$

B. Path Cost

The cost of a path is the sum of edge costs:

$$\text{Cost}(P_{ij}) = \sum_{k=1}^K \text{edge_cost}(p_{k-1}, p_k) \quad (63)$$

where:

$$\text{edge_cost}(p_{k-1}, p_k) = w_d \cdot \|p_k - p_{k-1}\| + w_z \cdot \max(0, z_k - z_{k-1}) \quad (64)$$

C. Optimal Path

The shortest feasible path is obtained via:

$$P_{ij}^* = \arg \min_P \text{Cost}(P) \quad (65)$$

subject to:

$$P \subset V \quad (66)$$

XII. ALGORITHM STRUCTURE

A. Online Algorithm Outline

The event-driven algorithm at the Command Center is:

Require: Global state $X(t)$

Ensure: Assignment $\mathbf{x}(t)$ and routes $\mathbf{P}(t)$

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1: while mission active do
2:   Wait for event trigger
3:   Update state:  $X(t) \leftarrow \text{getCurrentState}()$ 
4:   Update urgency:  $u_i(t) \leftarrow \text{decayUrgency}(u_i^0, t)$ 
5:   Update task queue:  $\mathcal{Q}(t) \leftarrow \text{orderQueue}(\mathcal{S}(t))$ 
6:   for each  $i \in \mathcal{Q}(t)$  do
7:     for each  $j \in \mathcal{D}(t)$  with  $A_j = 1$  do
8:       Compute path  $P_{ij}$  using 3D A*
9:       Compute  $E_{ij}^{\text{total}}$ 
10:      if  $E_{ij}^{\text{total}} + E_{\text{res}} \leq B_j$  then
11:        Mark  $(i, j)$  as feasible
12:        Compute cost  $C_{ij}$ 
13:      end if

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14: **end for**

15: **end for**

16: Solve assignment problem: $\mathbf{x}(t) = \arg \min \sum x_{ij} C_{ij}$

17: Dispatch assigned UAVs with routes $\mathbf{P}(t)$

18: Update state: $X(t) \leftarrow \text{applyDecision}(\mathbf{x}(t))$

19: **end while**

XIII. PERFORMANCE METRICS

A. Primary Metrics

$$\text{AvgWaitTime} = \frac{1}{N} \sum_{i=1}^N (t_i^{\text{arrive}} - t_i^{\text{det}}) \quad (67)$$

$$\text{SurvivorsReached} = \sum_{i=1}^N a_i \cdot [\text{responder arrived}] \quad (68)$$

$$\text{BatteryUtilization} = \frac{1}{M} \sum_{j=1}^M \frac{B_j^0 - B_j^{\text{final}}}{B_j^0} \quad (69)$$

B. Secondary Metrics

$$\text{EnergyEfficiency} = \frac{\text{SurvivorsReached}}{\sum E_{\text{consumed}}} \quad (70)$$

$$\text{ComputationTime} = t^{\text{solve}} - t^{\text{trigger}} \quad (71)$$

$$\text{UrgencyCoverage} = \frac{\sum u_i(t_i^{\text{arrive}})}{\sum u_i(t_i^{\text{det}})} \quad (72)$$

XIV. BASELINE MODEL: PASSIVE RECONNAISSANCE MODEL (PRM)

The PRM represents the conventional approach with no Responder UAVs:

- Survivors are detected by Scout UAVs but receive no intervention
- Survivors wait passively until Ground Rescue Team arrives
- No assignment or routing decisions are made

Performance is measured by:

$$\text{WaitTime}_{\text{PRM}} = t_i^{\text{ground}} - t_i^{\text{det}} \quad (73)$$

XV. SCOPE AND LIMITATIONS

A. Core Model Includes

- Survivor detection as exogenous input
- Responder UAV assignment and routing
- Battery-aware feasibility
- Time-dependent urgency
- Online, event-triggered decision making

B. Excluded (Future Work)

- Dynamic aftershock obstacles
- UAV-to-UAV collision avoidance
- Communication relay UAVs
- Hot-swap battery stations
- Detection uncertainty verification policies
- Recourse/reassignment after dispatch

XVI. CONCLUSION

This document presents a complete mathematical formulation of the Proactive Golden-Hour Bridge Model (PGBM) for post-earthquake UAV-assisted response. The model captures the essential dynamics of the system: survivor states with time-dependent urgency, Responder UAV states with battery constraints, and the online nature of detection arrivals and resource availability. The formulation separates feasibility (battery and routing constraints) from optimization (assignment and path selection), providing a clear foundation for algorithm design and simulation. The coupling between assignment and routing is explicitly defined through the cost function C_{ij} , which depends on both travel time and energy consumption.